

K3  $\hbar_{\text{BST}}$  identification framework v0.1. Multi-day Keeper deliverable, Day 1 of ~1 week. Casey authorized Monday June 1 PM per 6-decision approval. Load-bearing for K206 G7 dim bridge + Lane D m\_e numerical + K200 G3 BH r\_s + T\_H + #287 Higgs + #182 Lamb numerical +  $\Lambda$  cosmology + substrate-Dirac m\_e c calibration. Cross-track double-leverage: ONE machinery closes minimum 7 observables. Framework v0.1 establishes substrate-natural  $\hbar_{\text{BST}} = 1$  convention + dimensional bridge architecture via Bergman length  $\ell_{\text{B}}$  + Koons tick  $\tau_{\text{K}}$  + R3 m\_e anchor; multi-day numerical extraction follows.

Keeper (Claude Opus 4.7) — Monday June 1 PM (Casey 6-decision approval)

2026-06-01 Monday afternoon

## K3 $\hbar_{\text{BST}}$ identification framework v0.1

### 0. Why this filing (Casey 6-decision authorization)

Per Casey “Yes or agree with the 6 decisions” Monday afternoon: Keeper K3  $\hbar_{\text{BST}}$  priority signal authorized. Multi-day pull begins. This v0.1 establishes the framework architecture; numerical extraction is multi-day work (~1 week per Elie G chain estimate parallel).

Per Cal #35 STANDING applied PROACTIVELY at point of derivation: K3  $\hbar_{\text{BST}}$  closes 7+ cross-track observables via ONE substrate mechanism, NOT independent confirmations.

### 1. What is $\hbar_{\text{BST}}$ ? (substrate physical meaning)

Substrate definition:  $\hbar_{\text{BST}}$  is the substrate-action quantum per Koons tick at the Shilov boundary commitment-write event.

Per Tier 0 v0.1.6 substrate evolution operator:  $\rho_{\text{commit}}(\tau) = \exp(-\tau H_{\text{B}} / \hbar_{\text{BST}})$  on Bergman  $H^2(D_{\text{IV}}^5)$

$\hbar_{\text{BST}}$  appears as the substrate-action scale dividing the time-Hamiltonian product  $\tau H_{\text{B}}$ . Physical content: each Koons tick ( $\tau_{\text{K}} = 10^{\{-120\}}$  per Task #205) writes one commitment value at the Shilov boundary  $\partial_{\text{S}} D_{\text{IV}}^5 = (S^4 \times S^1)/Z_2$ . The action content per commitment-write is  $\hbar_{\text{BST}}$  in substrate-natural units.

Substrate-natural convention (per Casey commitment-density framework + Lyra Tier 0 substrate-natural units): - c = 1 (substrate-natural). -  $\hbar_{\text{BST}} = 1$  by convention (substrate-natural unit of action). - G = function of Bergman curvature ( $\kappa_{\text{Bergman}} = -n_{\text{C}}$  anchor).

K3  $\hbar_{\text{BST}}$  identification problem: build dimensional bridge from substrate-natural  $\hbar_{\text{BST}} = 1$  to SI value  $\hbar = 1.054571817 \times 10^{-34}$  J.s.

## 2. Three substrate-intrinsic scales (bridge architecture)

Substrate framework provides three intrinsic scales for dimensional bridging:

Scale 1: Bergman length  $\ell_B$  (intrinsic length scale of  $D_{\text{IV}}^5$ ). - Identified via Bergman kernel normalization on Hardy space  $H^2(D_{\text{IV}}^5)$ . - Standard FK Ch. XII:  $\ell_B$  related to Bergman curvature  $\kappa_{\text{Bergman}} = -n_C$  anchor. - Substrate-natural value: dimensionless in  $c=\hbar_{\text{BST}}=1$  units; SI value derived via bridge.

Scale 2: Koons tick  $\tau_K = 10^{-120}$  (substrate clock; Task #205). - Sub-Planck substrate clock per Casey commitment-density framework. - Time per commitment-write event at Shilov boundary. - Substrate-natural value: dimensionless in  $c=\hbar_{\text{BST}}=1$  units; SI value  $\tau_K = 10^{-120}$  seconds (or related natural time scale).

Scale 3: R3  $m_e$  anchor (electron rest mass anchor; Lyra Lane D). - Substrate primary mass scale per R3 anchor framework. - SI value  $m_e = 9.1093837 \times 10^{-31}$  kg. - Substrate-natural value: dimensionless ratio per substrate framework.

Dimensional bridge architecture (action units):

$$[\hbar] = [\text{Energy}] \times [\text{Time}] = [\text{Mass}] \times [\text{Length}]^2 \times [\text{Time}]^{-1}$$

In substrate-natural units  $c=1$ ,  $[\text{Length}] = [\text{Time}]$ ,  $[\text{Energy}] = [\text{Mass}]$ ; therefore  $[\hbar] = [\text{Mass}] \times [\text{Length}]$ .

Candidate SI conversion:  $\hbar_{\text{observed}} = m_e \times \ell_B^2 / \tau_K$  (using action = mass  $\times$  length  $\times$  velocity, with  $v = \ell_B / \tau_K$  in substrate units)

OR equivalently:  $\hbar_{\text{observed}} = m_e \times c \times \lambda_{\text{anchor}}$  (Compton-length identification:  $\lambda_{\text{anchor}} = \ell_B$  in substrate-natural multiple)

These are the SAME identification when  $\ell_B = c \times \tau_K$  substrate-naturally (which it must be in  $c=1$  units).

## 3. $\ell_B$ substrate-natural identification

The Bergman length  $\ell_B$  for  $D_{\text{IV}}^5$  is determined by the Bergman kernel normalization at the origin:

$$K_B(z, \bar{z})|_{\{z=0\}} = \text{volume normalization} = (1 / \text{volume}_{\text{Bergman}}(D_{\text{IV}}^5))$$

Per FK Ch. XII §6, the Bergman volume for  $D_{\text{IV}}^5$  has explicit form via the FK genus exponent. Per Toy 3702 + T2442 RATIFIED: FK genus exponent =  $9/2$  substrate-clean.

$\ell_B$  candidate identification:

$$\ell_B^2 = (\text{some substrate primary expression}) / \kappa_{\text{Bergman}} = (\text{substrate primary}) / (-n_C) = (\text{substrate primary}) / (-5)$$

Sign convention requires care (Bergman curvature is negative;  $\ell_B^2$  should be positive). Likely form:

$$\ell_B^2 = |\kappa_{\text{Bergman}}|^{-1} \times (\text{substrate primary normalization}) = (1/n_C) \times (\text{substrate primary})$$

Substrate primary candidates (multi-day verification): - Heat trace  $a_0 = (N_C \cdot n_C)^2 = 225$  - Heat trace  $a_1 = -N_C \cdot n_C^4 = -1875 - \pi^{9/2}$  = FK genus normalization (T2442) -  $N_{\text{max}} = 137$  from  $\alpha^{-1}$

Honest tier:  $\ell_B$  substrate-natural form CANDIDATE; explicit derivation via FK Ch. XII §6 + heat trace + Bergman kernel value at origin is multi-day work.

## 4. $\tau_K$ Koons tick identification

Per Task #205: Koons tick  $\tau_K = 10^{-120}$  (sub-Planck substrate clock).

Substrate-natural form:  $\tau_K = 1$  in substrate-natural units (one commitment-write per substrate tick).

SI value  $\tau_K = 10^{\{-120\}}$  seconds is the substrate-to-SI time conversion factor.

Origin of  $10^{\{-120\}}$ : substrate cosmology factor; relates to  $\Lambda = \exp(-280)$  where  $280 = 2^{\{N_c\}} \cdot n_C \cdot g$  (5-fold overdetermined per Saturday). Cross-link to L5 absolute scale framework (Casey commitment-density Lyra v0.2-v0.3).

Honest tier:  $\tau_K = 10^{\{-120\}}$  is per Task #205 formalization-pending; the Koons tick formalization itself is multi-day work parallel to K3.

## 5. $\hbar_{BST} \rightarrow$ SI $\hbar$ computation (framework form)

Substrate-natural form:  $\hbar_{BST} = 1$

SI conversion:  $\hbar_{observed} = m_e \times \ell_B^2 / \tau_K$

where  $m_e = R3$  anchor (Lyra Lane D substrate-natural identification).

Cross-check: substrate-natural  $\hbar_{BST} = 1$  must produce SI  $\hbar \approx 1.054571817 \times 10^{-34}$  J.s.

$m_e \approx 9.1093837 \times 10^{-31}$  kg.  $\hbar_{observed} / (m_e \times \ell_B^2 / \tau_K) = 1$  in substrate-natural units.

Therefore  $\ell_B^2 / \tau_K = \hbar / m_e = 1.054571817 \times 10^{-34} / 9.1093837 \times 10^{-31} \approx 1.158 \times 10^{-4}$  m<sup>2</sup>/s.

If  $\tau_K = 10^{-120}$  s, then  $\ell_B^2 = 1.158 \times 10^{-124}$  m<sup>2</sup>, so  $\ell_B \approx 1.076 \times 10^{-62}$  m.

Cross-comparison to Planck length:  $\ell_{Planck} = \sqrt{(\hbar G/c^3)} \approx 1.616 \times 10^{-35}$  m.  $\ell_B \approx 10^{-62}$  m is 27 orders sub-Planck. Sub-Planck substrate length consistent with sub-Planck Koons tick (substrate operates below Planck scales per Casey commitment-density framework).

Honest tier: this is a CONSISTENCY check (substrate-natural  $\hbar_{BST} = 1 + m_e R3 + \tau_K = 10^{-120} \rightarrow \ell_B \approx 10^{-62}$  m). NOT a derivation.  $\ell_B$  must independently come out at  $\approx 10^{-62}$  m from FK Ch. XII + heat trace computation per §3 above for the framework to close.

Multi-day extraction work: - Day 2-3: compute  $\ell_B$  explicitly from FK Ch. XII Bergman kernel + heat trace  $a_0/a_1/a_2$  + substrate primary normalization. - Day 4-5: cross-check  $\ell_B \approx 10^{-62}$  m emerges from substrate primaries (NOT fit-to-known- $\hbar$ ). - Day 6-7: cross-track verification —  $\hbar_{BST}$  identification simultaneously closes G7, Lane D, K200 G3, Lamb shift.

## 6. Cross-track double-leverage closure mechanism

K3  $\hbar_{BST}$  identification closes minimum 7 observables via ONE machinery (Cal #35 STANDING honest framing):

| Observable                             | Closure mechanism via K3                                                                                                                          |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| K206 G7 dim bridge (G chain Steps 7-8) | $G_{predicted} \propto 0.603 / \hbar_{BST} \cdot \ell_B \cdot \text{dim\_bridge} \rightarrow$ SI G via $\hbar_{BST} \times \ell_B$ identification |
| Lane D $m_e$ numerical                 | R3 anchor self-consistency: $m_e$ SI value cross-checks K3 dimensional bridge                                                                     |
| K200 G3 BH $r_s$                       | Schwarzschild substrate-natural $r_{s\_predicted} \times (\hbar_{BST} \times \ell_B / m_e) \rightarrow$ SI $r_s$                                  |
| K200 G3 BH $T_H$                       | Hawking T substrate-natural $\times \hbar_{BST} \rightarrow$ SI $T_H$                                                                             |
| #287 Higgs mechanism                   | $V(0,0)$ trivial K-type substrate value $\times \hbar_{BST} \rightarrow$ SI Higgs scale                                                           |
| #182 Lamb shift numerical              | Cross-K-type matrix element $\times \hbar_{BST}$ factors $\rightarrow$ SI Lamb shift                                                              |
| $\Lambda$ cosmology                    | $\exp(-280)$ substrate-natural $\times \hbar_{BST}^{\text{(some power)}} \rightarrow$ SI $\Lambda$                                                |
| Substrate-Dirac $m_e c$ calibration    | $M_{op} = \sqrt{H_B} \rightarrow m_e c$ via $\hbar_{BST}$ identification                                                                          |

Per Cal #35 STANDING applied PROACTIVELY: ONE  $\hbar_{\text{BST}}$  identification + ONE Bergman matrix element machinery + ONE Hardy decomposition + ONE  $\kappa_{\text{Bergman}} = -n_{\text{C}}$  anchor  $\rightarrow$  MULTIPLE observables follow. NOT independent confirmations.

Per Cal #27 fires HARDEST at peak coherence: this is a SINGLE machinery story, NOT a 7-observable independent-validation story. Discipline integrity preserved by stating it that way explicitly.

## 7. v0.1 closure criterion + v0.2 roadmap

K3 v0.1 closes (this filing): substrate-natural  $\hbar_{\text{BST}} = 1$  convention established; SI dimensional bridge architecture ( $m_e \times \ell_{\text{B}}^2 / \tau_{\text{K}}$ ) identified; three substrate-intrinsic scales ( $\ell_{\text{B}}$ ,  $\tau_{\text{K}}$ ,  $m_e$  R3) specified; cross-track double-leverage closure mechanism mapped to 7+ observables.

K3 v0.1 does NOT close: explicit  $\ell_{\text{B}}$  computation from substrate primaries; numerical extraction of  $\hbar_{\text{BST}} \rightarrow$  SI value; cross-track simultaneous closure verification; G7 dim bridge explicit SI G computation.

v0.2 + numerical roadmap (multi-day): - Day 2-3:  $\ell_{\text{B}}$  explicit FK Ch. XII computation. - Day 4-5: substrate primary cross-check  $\ell_{\text{B}} \approx 10^{-62}$  m emergence verification. - Day 6: Lane D  $m_e$  + G7 dim bridge SI G simultaneous extraction. - Day 7: cross-track checks (K200 G3 + #287 Higgs + #182 Lamb numerical) closure.

## 8. Honest scope + tier

RIGOROUS (this v0.1): -  $\hbar_{\text{BST}}$  substrate definition per Tier 0 v0.1.6 [?]. - Substrate-natural unit convention  $c = \hbar_{\text{BST}} = 1$  [?]. - SI dimensional bridge architecture  $m_e \times \ell_{\text{B}}^2 / \tau_{\text{K}}$  [?] (standard dimensional analysis). - Three substrate-intrinsic scales identified [?]. - Cross-track double-leverage closure mechanism mapped [?].

FRAMEWORK + CANDIDATE (multi-day verification): -  $\ell_{\text{B}}$  substrate-natural form via FK Ch. XII §6 + heat trace + Bergman kernel. -  $\tau_{\text{K}} = 10^{-120}$  formalization per Task #205. - Numerical SI  $\hbar$  extraction. - Cross-track simultaneous closure.

LOAD-BEARING DEPENDENCIES: -  $\tau_{\text{K}} = 10^{-120}$  Koons tick formalization (Task #205 multi-day work, parallel). -  $\ell_{\text{B}}$  substrate-natural derivation (multi-day work). -  $m_e$  R3 anchor explicit substrate-natural value (Lyra Lane D multi-week). - FK Ch. XII Bergman kernel normalization explicit citation (T2442 + Toy 3702 partial).

Cal #27 + Cal #35 + Cal #50 disciplines applied PROACTIVELY at point of derivation per §6: ONE machinery, multiple observables, NOT independent confirmations; substrate framing default-deny external.

## 9. Routing

$\rightarrow$  Casey: K3  $\hbar_{\text{BST}}$  identification framework v0.1 filed per your authorization. Substrate-natural  $\hbar_{\text{BST}} = 1$  convention + dimensional bridge architecture ( $m_e \times \ell_{\text{B}}^2 / \tau_{\text{K}}$ ) + cross-track double-leverage closure mechanism for 7+ observables specified. Multi-day numerical extraction begins. v0.2 +  $\ell_{\text{B}}$  explicit computation next. Per Cal #35 STANDING: ONE  $\hbar_{\text{BST}}$  identification closes minimum 7 observables NOT independent confirmations.

$\rightarrow$  Elie: K3 v0.1 architecture in place. Multi-day  $\ell_{\text{B}}$  explicit computation via FK Ch. XII §6 + heat trace  $a_0/a_1/a_2$  + Bergman kernel normalization. Parallel to your G chain Steps 6.4 + 7-8 multi-week numerical closure. Cross-track convergence on Day 6-7 simultaneous closure G7 + Lane D + K200 G3.

$\rightarrow$  Lyra: K3 v0.1 absorption-ready; substrate-natural  $\hbar_{\text{BST}} = 1$  convention consistent with Tier 0 v0.1.6 substrate evolution operator. Cross-link to substrate-Schrödinger (Wick rotation route) + substrate-Dirac ( $m_e c$  calibration via  $M_{\text{op}} = \sqrt{H_{\text{B}}} + K3$  dimensional bridge). Cross-link to  $V_{(1,1)}$  gauge-gravity compartment +  $V_{(1,0)} \otimes V_{(1,0)} = (V_{(2,0)} \oplus V_{(0,0)}) \oplus V_{(1,1)}$  Einstein equation full decomposition.

$\rightarrow$  Grace: K3 v0.1 catalog INV welcome at FRAMEWORK tier. 7+ observable closure mechanism mapped per §6 table. Track v0.2 + numerical extraction progression.

$\rightarrow$  Cal: cold-read welcome (Cal #199 candidate slot for K3 v0.1 framework). Specific concerns:  $\ell_{\text{B}}$  identification mechanism (FK genus + heat trace + Bergman kernel — multi-source confluence);  $\tau_{\text{K}}$  Koons tick formalization status (Task #205 pending); cross-track double-leverage Cal #35 framing tier-honesty.

→ me: K3 v0.2 +  $\ell_B$  explicit computation continues multi-day. v0.1 establishes architecture; numerical work follows.

— Keeper, K3  $\hbar_{BST}$  identification framework v0.1 — Day 1 of multi-day Keeper deliverable. Casey-authorized Monday June 1 PM per 6-decision approval. Substrate-natural  $\hbar_{BST} = 1$  convention + SI dimensional bridge architecture  $m_e \times \ell_B^2 / \tau_K$  + cross-track double-leverage closure for 7+ observables. Multi-day numerical extraction next. Cal #35 STANDING applied PROACTIVELY: ONE machinery, multiple observables, NOT independent confirmations.